Core Java Study Plan Day 5

Interview Questions

1. What is Object-Oriented Programming (OOP)?

2. What are the main concepts of OOP in Java?

3. What is a class in Java?

4. What is an object in Java?

5. What is the difference between a class and an object?

6. How do you create an object of a class in Java?

7. What is the purpose of a constructor in Java?

8. What is the difference between a constructor and a method?

9. Can a constructor have a return type?

10. What happens if you do not create a constructor in a Java class?

11. What is a default constructor?

12. What is a parameterized constructor?

13. Can a class have multiple constructors?

14. What is constructor overloading?

15. What is a method in Java?

16. What is the difference between instance methods and static methods?

17. What is the use of the this keyword in Java?

18. How is this used to differentiate between instance variables and local variables?

19. Can this keyword be used inside a static method? Why?

20. How can one constructor call another constructor of the same class using this?

MCQ Questions

1. Which of the following is used to define a class in Java?

A. object

B. class

C. new

D. method

2. Which keyword is used to create an object in Java?

A. class

B. object

C. new

D. create

3. Which of the following represents an instance of a class?

A. Method

B. Constructor

C. Object

D. Variable

4. Which special method is automatically called when an object is created?

A. Main method

B. Constructor

C. Static method

D. Getter method

5. Which of the following is NOT a valid return type for a constructor?

A. int

B. void

C. String

D. None of these

6. Which keyword refers to the current object?

A. super

B. this

C. current

D. object

7. Which of the following allows multiple constructors in the same class with different parameters?

A. Method overriding

B. Constructor overloading

C. Method hiding

D. Encapsulation

8. What is the main purpose of a method in Java?

A. To define a package

B. To perform a specific task

C. To create a class

D. To allocate memory

9. Which keyword can be used to call another constructor of the same class?

A. super

B. this

C. new

D. static

10. Which of the following can contain instance variables and methods?

A. Class

B. Constructor

C. Object

D. Package

Puzzles

1. You have three boxes. One contains only apples, one contains only oranges, and one contains both apples and oranges. All three boxes are incorrectly labelled. You may pick only one fruit from one box without looking inside. From which labelled box should you pick, and how can you correctly label all three boxes?

2. You have three switches outside a closed room. Inside the room there is one ordinary bulb. Only one switch controls the bulb. You may operate the switches as much as you want, but you may enter the room only once. How will you identify the correct switch?

3. Two ropes are available. Each rope takes exactly 60 minutes to burn completely, but they do not burn at a uniform rate. Using only these two ropes and a lighter, how can you measure exactly 45 minutes?

4. A farmer needs to take a wolf, a goat and a basket of vegetables across a river. The boat can carry the farmer and only one of the three items at a time. The wolf cannot be left alone with the goat, and the goat cannot be left alone with the vegetables. How can the farmer get everything safely across?

5. You have 12 identical-looking coins. One coin is counterfeit and is either heavier or lighter than the others. Using a balance scale exactly three times, how can you identify the counterfeit coin and determine whether it is heavier or lighter?

Aptitude Questions

APTITUDE TEST – SET 5

Percentage, Profit & Loss, Ratio & Proportion | 20 Questions

1. In a company, 40% of employees are women. Of the women, 30% have completed a certification, while 20% of the men have completed it. If 100 employees have completed the certification, how many employees are there in total?

A) 400 B) 450 C) 500 D) 550

2. A test has 80 questions. A candidate answers 75% of the first 40 questions correctly and 60% of the remaining questions correctly. What percentage of the entire test did the candidate answer correctly?

A) 65% B) 67.5% C) 70% D) 72.5%

3. The price of a product is increased by 20%. A retailer then gives a discount so that the final selling price is 4% higher than the original price. What is the discount percentage on the increased price?

A) 10% B) 12% C) 13 1/3% D) 15%

4. A candidate's marks are 25% higher than those of another candidate. If the first candidate scored 300 marks, how many marks did the second candidate score?

A) 225 B) 240 C) 250 D) 260

5. A factory reduces its electricity consumption by 15% in one month and by another 10% the following month. If the original consumption was 40,000 units, how many units are saved over the two months?

A) 9,000 B) 9,400 C) 9,600 D) 10,000

6. A number is 20% less than another number. If the smaller number is increased by 25, it becomes equal to 90% of the larger number. Find the larger number.

A) 100 B) 125 C) 150 D) 200

7. In an election, 10% of registered voters did not vote. Of the votes cast, candidate A received 60% and candidate B received the remaining votes. If B received 18,000 votes, how many voters were registered?

A) 45,000 B) 50,000 C) 55,000 D) 60,000

8. A trader buys an article at 20% below its marked price and sells it at 10% below the marked price. What is his profit percentage on the cost price?

A) 10% B) 12.5% C) 15% D) 20%

9. A shopkeeper sells two articles for ₹1,500 each. He gains 25% on one and loses 25% on the other. What is the total loss?

A) ₹150 B) ₹200 C) ₹240 D) ₹300

10. A trader sells an article for ₹2,760 after earning 15% profit. If he wants to earn 25% profit, by how much should he increase the selling price?

A) ₹200 B) ₹220 C) ₹240 D) ₹260

11. A product is sold after successive discounts of 15% and 10%. If the marked price is ₹4,000, what is the selling price?

A) ₹3,000 B) ₹3,060 C) ₹3,100 D) ₹3,200

12. A dealer purchases an article for ₹2,400. He sells it at a profit of 12.5%. If he had sold it for ₹150 less, what would have been the profit percentage?

A) 5% B) 6.25% C) 7.5% D) 8.33%

13. A seller marks an article 60% above cost price and allows a discount of 25%. If the selling price is ₹2,400, find the cost price.

A) ₹1,800 B) ₹2,000 C) ₹2,100 D) ₹2,400

14. A dishonest shopkeeper uses a weight of 800 g instead of 1 kg but charges the price of 1 kg. What is his gain percentage?

A) 20% B) 22.5% C) 25% D) 30%

15. A:B:C = 2:3:7. If C gets ₹1,000 more than A, what is the total amount?

A) ₹2,000 B) ₹2,400 C) ₹2,500 D) ₹3,000

16. The ratio of boys to girls in a college is 7:5. If 24 more girls join, the ratio becomes 7:6. How many boys are there?

A) 144 B) 168 C) 192 D) 216

17. A:B = 4:7. If 8 is subtracted from each number, the ratio becomes 2:5. Find the smaller number.

A) 10 B) 12 C) 16 D) 20

18. Three partners invest in the ratio 2:3:5. If the first partner invests for 12 months, the second for 8 months and the third for 6 months, what is their profit-sharing ratio?

A) 4:4:5 B) 1:1:1 C) 2:2:3 D) 3:2:5

19. A mixture contains milk and water in the ratio 5:3. If 16 litres of water are added, the ratio becomes 5:7. Find the original quantity of milk.

A) 20 L B) 25 L C) 30 L D) 40 L

20. The speeds of two trains are in the ratio 5:7. If the faster train covers 420 km in 6 hours, how far will the slower train travel in 5 hours?

A) 200 km B) 220 km C) 250 km D) 300 km